



# HAMMARD UNIVERSITY BANGLADESH

**Faculty of Science, Engineering and Technology**  
**Department of Computer Science and Engineering**  
**CSE 415: Communication Engineering**

Batch: 22<sup>nd</sup>

Date: 10/09/2025

Final Examination

Semester: Spring 2025

Time: 2 Hours

Total Marks: 40

Answer any five questions:

5 × 8 = 40

1. (a) Draw a figure to prove the relationship between STS and STM. 4  
(b) Illustrate STS-3 frame with its overhead. 4
2. (a) List the tasks performed by the signaling system in Telephone Network. 4  
(b) Describe the SS7 services and its relation to the telephone network. 4
3. (a) In ATM, write the relationship between TPs, VPs, and VCs. 4  
(b) Draw and explain Wireless LAN (IEEE 802.11) architecture. 4
4. (a) Explain why SNR in a wireless LAN is normally lower than SNR in a wired LAN? 4  
(b) Compare Piconet with Scatternet in the Bluetooth architecture. 4
5. (a) What is a Footprint? Explain the purposes of GPS. 4  
(b) Briefly describe the operations of the cellular telephony. 4
6. (a) Write the relationship between the VAN Allen belts with Satellites. 4  
(b) Describe the function of the CDMA in IS-95. 4
7. (a) Explain the differences between a fixed WiMAX and a mobile WiMAX. 4  
(b) Write the differences between a hard handoff and a soft handoff. 4

TEAP  
step



**Faculty of Science, Engineering and Technology**  
**Department of Computer Science and Engineering**  
**CSE 417: Compiler Design**

Batch: 22<sup>nd</sup>

Semester: Spring 2025

Date: 17/09/2025

**Final Examination**

Time: 2 hours

Total Marks: 40

Answer any five of the following questions.

Marks: 5 x 8=40

1. (a) Define Context-free grammar and Recursive grammar. 2  
 (b) Write the differences between SDD and SDT. 3  
 (c) State the summary of recursive descent parsing of a compiler. 3
  
2. (a) Write down about ambiguous grammar and unambiguous grammar. 2  
 (b)  $E \rightarrow + FT$  3  
 $T \rightarrow * GH$  3  
 Make synthesized on E node and make Inherited on G node.  
 (c)  $S \rightarrow AA$   
 $A \rightarrow aA/b$   
 Show whether the above grammar is accepted by LR(0) parser or not.
  
3. (a) Write the rules for L-attributed grammars. 2  
 (b) Make the table for Quadruples and Triples for the expression :  $k = c * - b + c * - c + a$  6
  
4. (a) State the differences between an annotated parse tree and a dependency graph. 2  
 (b) Draw an annotated tree for :  $(9+8*(7+6))*4n$  . 6
  
5. (a) State the rules for Quadruples. 3  
 (b) Construct a DAG for this expression:  $a + a * (b - c) + (b - c) * d$ . 5
  
6. (a) Write three address code for the assignment:  $a = b * - c + b * - c$ . 2  
 (b)  $S \rightarrow Aa/bAc/dc/bda$  6  
 $A \rightarrow d$   
 Show whether the grammar is accepted by an SLR(1) parser or not.
  
7. (a)  $S \rightarrow S \# A / A$  4  
 $A \rightarrow A \& B / B$   
 $B \rightarrow id$   
 Given String:  $(10 \& 4) \# 6$   
 Draw an annotated tree from the given grammar and show the output for the given string.  
 (b)  $E \rightarrow E + T$  4  
 $E \rightarrow T$   
 $T \rightarrow T * F$   
 $T \rightarrow F$   
 $F \rightarrow digit$   
 Draw the dependency graph for:  $10 * 4 + 3 * 7 + 8$ .



# HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE471: Computer and Network Security

Final Examination

Batch: 22<sup>nd</sup>

Date: 15/09/2025

Semester: Spring 2025

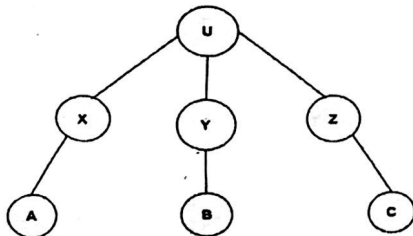
Time: 2 Hours

Answer any **five** questions from the followings.

Total Marks: 40

1. a) What is the CIA triad? Explain. 2  
b) Define non-repudiation and authentication. How do they relate to each other? 3  
c) How does ensuring data integrity contribute to authentication? Explain. 3
2. a) What is SQL injection? Explain. 2  
b) Differentiate between weakness, vulnerability and attack in the context of cryptography. 3  
c) Decipher the following text using Rail-Fence cipher where the depth is 2. 3  

**UIESTNVRIY**
3. a) What is the primary purpose of the Oakley Key Determination Protocol and how does it enhance Diffie-Hellman key exchange? 4  
b) What is SSL and what is its primary purpose? Explain the SSL handshaking protocol and illustrate the key steps involved in establishing a secure connection. 4
4. a) What is a digital certificate? 1  
b) Briefly explain the concept of Public Key Infrastructure (PKI) and mention its components. 3  
c) In an X.509 authentication framework, entities U, X, Y, Z, A, B, and C follow the Chain of Certificates rule. Given the certificate hierarchy as shown, explain how entity A can securely communicate with entity C? 4



5. a) Explain the use of rainbow table in Birthday Attack. 2  
b) What is cryptographic salt? Explain why it is used in cryptographic applications. 2  
c) Explain how PGP (Pretty Good Privacy) works, including its cryptographic processes. Provide two distinct scenarios where PGP is used, and include necessary diagrams or illustrations to support your explanation. 4
6. a) Explain the role of cryptography in the UNIX password scheme, including how cryptographic techniques are used to securely store and verify user passwords. 4  
b) Explain why MIME and S/MIME protocols were introduced after SMTP. 4
7. c) Illustrate and explain the steps involved in the Kerberos authentication protocol when a client requests access to a local server within a network. 8



# HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology  
Department of Computer Science and Engineering  
CSE413: Computer Graphics

Batch: 22<sup>nd</sup>

Semester: Spring 2025

Date: 13/09/2025

Final Examination

Time: 2 hours

Total Marks: 40

Answer any five questions from the following:

5x8= 40

- 1 (a) Define Ellipse and state the properties of ellipse. 4  
(b) Consider an ellipse with a semi-major axis of  $a=12$  units and a semi-minor axis of  $b=8$  units. Use the midpoint ellipse algorithm to determine the first few points on the ellipse starting from the center at  $(0, b)$  and moving towards the first quadrant. 4
- 2 (a) Explain four-way symmetry. 3  
(b) Briefly explain Conceptual model of 3D viewing process. 5
- 3 (a) Describe two types of transformation in Computer Graphics. 2  
(b) Consider a point  $P(1,2,3)$  in 3D. Apply the following transformations to each point: 6
  - i. Translate the point  $P(1, 2, 3)$  by  $T_x=2$ ,  $T_y=3$ , and  $T_z=4$ .
  - ii. Rotate the point  $P(1, 2, 3)$  by  $\theta=90^\circ$  anti-clockwise around the  $z$ -axis.
  - iii. Scale the point  $P(1, 2, 3)$  by  $S_x=2$ ,  $S_y=3$ , and  $S_z=4$ .
- 4 (a) Explain Phong model. 2  
(b) Convert the RGB color ( $R=120$ ,  $G=200$ ,  $B=150$ ) to the CMY and YIQ color model. 6
- 5 (a) State the advantages and disadvantages of orthographic projection. 2  
(b) A 3D point  $P(25.0, 15.0, -600.0)$  is projected onto a projection plane whose center is at  $(0.0, 0.0, -400.0)$ . The center of projection (COP) is located at  $(50.0, 30.0, 0.0)$ . Determine the coordinates of the projected point on the projection plane using the general-purpose projection matrix approach. 6
- 6 (a) Determine whether the following lines are accepted/rejected/partial-using Cohen Sutherland line clipping algorithm. 4  
Given  $(-250,-200)$  to  $(250,200)$  be the clip region.
  - i.  $(-100, -220)$  to  $(300, -210)$ .
  - ii.  $(-250, 200)$  to  $(250, -200)$ .  
(b) A triangle has vertices  $P(1, 1)$ ,  $Q(4, 1)$ , and  $R(2, 4)$ . Clip this triangle against a rectangular clipping window  $x_{\min}=2$ ,  $y_{\min}=2$ ,  $x_{\max}=5$ , and  $y_{\max}=3$  using the Sutherland-Hodgman algorithm. List the clipped polygon vertices. 4



- 7
- (a) State the differences between orthographic and oblique projection. 2
  - (b) A point  $P(3, 4)$  is translated by  $T_x=2$  and  $T_y=5$ . Find the new coordinates using homogeneous transformation. 3
  - (c) Rotate the point  $P(2, 3)$  by  $90^\circ$  counterclockwise using homogeneous coordinates. 3



# HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology  
Department of Computer Science and Engineering  
CSE 411: Artificial Intelligence

Batch: 22<sup>nd</sup>

Semester: Spring 2025

Date: 08/09/2025

Final Examination

Time: 2 hour

Total Marks: 40

Answer any five questions from the following.

5x8= 40

1. (a) Given that a  $\vec{v} = [3, 4, 5, 6, 7, 8]$ , 2  
1. Calculate the magnitude of vector  
2. Compute the length of  $m \cdot \vec{v}$ , where  $m = 10$
- (b) Given that  $\vec{v} = [6, 7, 8]$  3  
1. Compute  $|\vec{v}|$ ,  
2. Compute angles i.e.  $\alpha, \beta, \gamma$
- (c) Define true or false 3  
1.  $2 + 3i$  is a vector  
2.  $[1, 2, 3, 4]$  is a four-dimensional scalar.  
3.  $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cdot \sin \theta$
2. (a) Given that,  $A \cdot \vec{v} = \lambda \cdot \vec{v}$  2  
1. What is the  $\vec{v}$  of A?  
2. What is  $\lambda$ ?
- (b) Given that  $A = [1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9]$  3  
1. Find the  $\det(A)$ .  
2. Find  $A^{-1}$
- (c) Define True or false 3  
1. if  $A = A^T$ , then is A orthogonal matrix.  
2. Matrix Multiplication is non-commutative.  
3. Matrix scalar multiplication is associative.
3. (a) Prove that,  $\frac{d}{dx}(f(x)) = \int f'(x)dx$ , where  $f(x) = 5x^2 + 3x$  2
- (b) Given that  $f(x) = x^2 + 5x + 10$  and  $g(x) = x^3 + 2x + 1$  3  
1. Calculate:  $\frac{d}{dx}(f(x) + g(x))$   
2. Calculate:  $\frac{d}{dx}(f(x) - g(x))$
- (c) Given that  $f(x) = 2x + 5$  and  $g(x) = 3x^2 + 2$  3  
1. Calculate:  $\int (f(x) + g(x))dx$   
2. Calculate:  $\int (f(x) - g(x))dx$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

4. (a) Given that Probability  $P(A) = 0.5$  and  $P(B) = 0.5$ .  
What will be the joint probability ?
- (b) Give that,

| Day | Sky Condition | Rain |
|-----|---------------|------|
| 1   | Sunny         | No   |
| 2   | Cloudy        | No   |
| 3   | Rainy         | Yes  |
| 4   | Sunny         | No   |
| 5   | Cloudy        | Yes  |
| 6   | Rainy         | Yes  |
| 7   | Sunny         | No   |
| 8   | Cloudy        | No   |
| 9   | Rainy         | Yes  |
| 10  | Sunny         | No   |

Calculate the probability of rain, using Bayes theorem.

- (c) True or false ? "Probability can be negative"

5. (a) Given that,  $A = [4, 4, 3.5, 3, 3.5, 3.5, 2.5, 0, 2.5, 3.5, 0]$   
Calculate:
1. Mean,
  2. Median
  3. Mode

- (b) Given that, Contingency table

| Gender | Pass | Fail | Total |
|--------|------|------|-------|
| Male   | 3    | 5    | 8     |
| Female | 7    | 5    | 12    |
| Total  | 10   | 10   | 20    |

$H_0 =$  Gender is related to pass or fail. Test that hypothesis and conclude.

- (c) True or False: t-test uses mean for hypothesis testing.

6. (a) Write a match-case statement using python.
- (b) Write python functions for:

1. Mean
2. Mode
3. Median

- (c) True or False ?

1. "else-if is available in python"
2. "For real-time AI model compiler based language is best regarding timely response."

3. "Java needs Interpreter"

7. (a) What is relation operator in python. Write a complete list of Comparison operators. 2
- (b) Calculate: 4
1.  $5 \& 3 = ?$
  2.  $5 | 3 = ?$
  3.  $5 \text{ and } 5 = ?$
  4.  $5 \text{ or } 3 = ?$
- (c) True or False?: 2
1. Python is not an object oriented language.
  2. Python has not the best eco-system for AI.